Campus Maintenance System

Controlled, Auditable Complaint-to-Closure Relational Database

3NF Normalized | 14 Entities | Status Control | MySQL Backend

DBMS Project 41 — Review 1

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System Necessity

Lost Requests
Centralized unique request tracking.

Informal Assignment
Strict capacity limits per technician.

No Cost Tracking
Structured material and labor logs.

Poor Visibility
Real-time status and aging reports.

Objectives & Scope

Primary Objectives

- 14-entity relational model.
- Strict 3NF normalization.
- Database-level constraint enforcement.
- Analytical reporting views.
- Java front-end integration.

In Scope

- Request priority tracking.
- Technician task assignment.
- Material consumption logs.

Out of Scope

- Payroll processing.
- Procurement workflows.
- Campus ERP integration.

System Roles



Requester

Issue tracking and status monitoring.



Facility Admin

Resource assignment and ticket closure.



Technician

Labor and material consumption recording.



Management

Performance and cost analytics.

Core Requirements

1. Log unique requests.
2. Define SLA priorities.
3. Assign technical staff.
4. Log work & materials.
5. Track status flow.
6. Capture closure feedback.

Search & Analytics: Aging, response times, cost per building, technician load.

Conceptual ER Model

14 Entities

Building, Room, Asset

User, Technician

Category, Priority

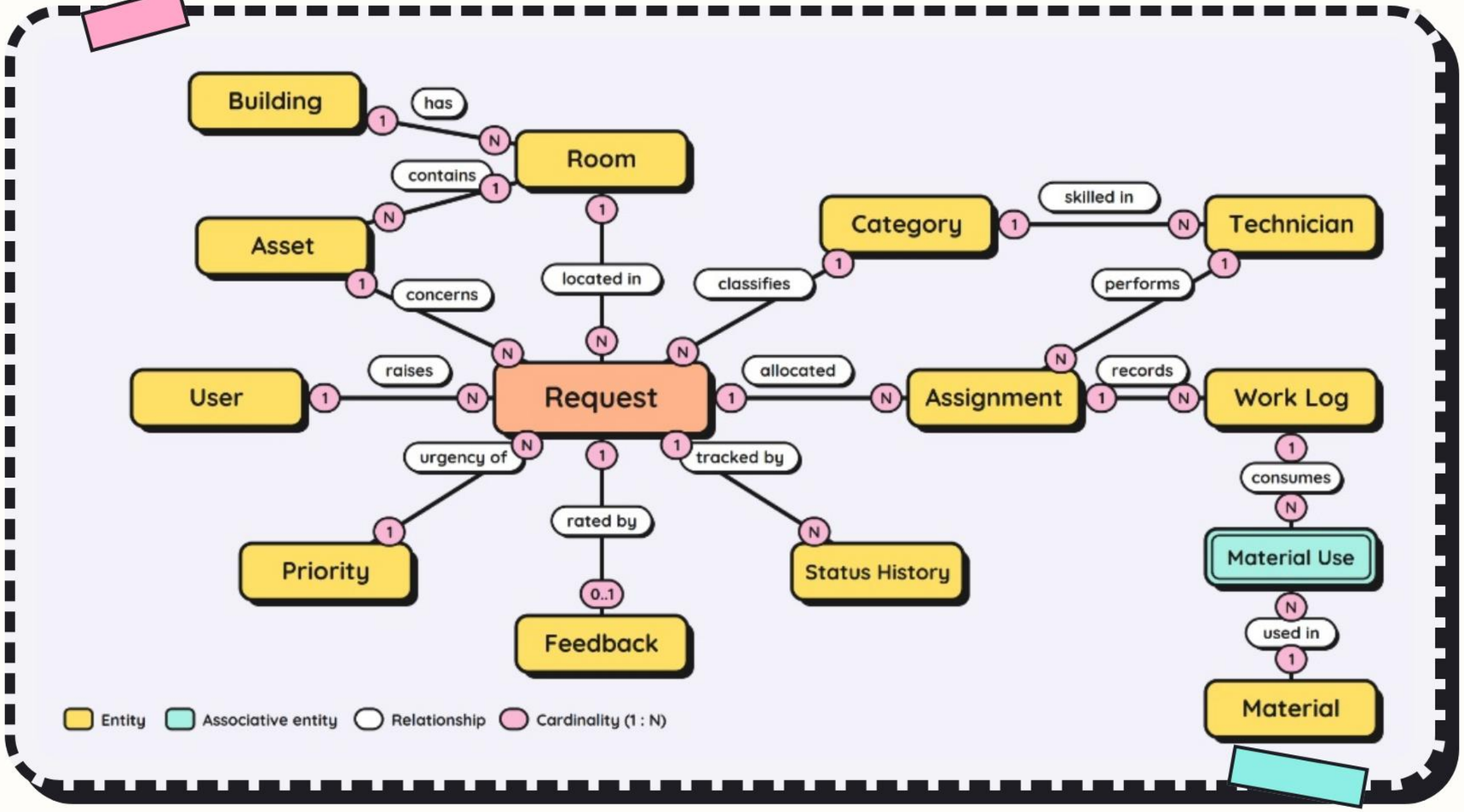
Request, Assignment

Work Log, Material

Material Use

Status History

Feedback



☆ **Request is the central hub. Material Use resolves M:N Work Log relationships.**

3NF Relational Schema

building
- **building_id**
- name

room
- **room_id**
- **building_id**
- room_no

asset
- **asset_id**
- **room_id**
- name

app_user
- **user_id**
- name
- email
- role

category
- **category_id**
- name

priority
- **priority_id**
- name
- sla_hours

technician
- **tech_id**
- name
- **category_id**

material
- **material_id**
- name
- unit_cost

request
- **request_id**
- **user_id,**
- **room_id...**
- status, dates

assignment
- **assign_id**
- **req_id, tech_id**

work_log
- **log_id**
- **assign_id**
- hours

material_use
- **use_id**
- **log_id, mat_id**
- qty

status_hist
- **hist_id**
- **req_id**
- stat changes

feedback
- **request_id**
- rating
- comments

Primary Key **Foreign Key**

Integrity & Constraints

3NF Verification

1NF Strict atomic attributes.

2NF Zero partial dependencies.

3NF Zero transitive dependencies.

Business Rules

- Unique request identifiers.
- Mandatory closure remarks.
- Strict non-negative constraints.
- Valid status transitions.
- Technician load limits.
- Post-closure feedback limit.

Project Roadmap

1

Review 1

(Current)

ER model and 3NF schema.

2

Review 2

Constraints, triggers, views.

3

Review 3

Java application demo.

Thank you! Questions?